

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: 2.2 CHEMICAL FAMILIES – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle. Remember Mama Wanjiru, the food seller in Mombasa, who asked why her aluminium sufuria is safe while chlorine in her tap water is poisonous as a gas. Use the evidence you collected in Lessons 1 to 8, the class family model and the Driving Question Board (DQB) to answer each question in the space below it. Write in full sentences. For each answer, use a claim (what you think), evidence (data or observations from our lessons) and reasoning (the idea of outer electrons and position in the periodic table that links them). Use the words element, compound, group, period and chemical family correctly. Write balanced equations with $\rightarrow$ where asked. Use these data when you need them. Sodium (Na): density 0.97 g/cm³, melting point 98 C. Magnesium (Mg): density 1.74 g/cm³, melting point 650 C. Aluminium (Al): density 2.70 g/cm³, melting point 660 C. Copper (Cu): density 8.96 g/cm³, melting point 1085 C, excellent conductor. Lead (Pb): density 11.34 g/cm³, melting point 327 C. Water has a density of 1.00 g/cm³. Chlorine is added to Mombasa water at about 0.5 mg/L.

Part 1: Why is sodium dangerous alone?

In the demonstration, Mr. Otieno dropped a pea-sized piece of sodium into water behind a safety screen. (a) Describe what you saw and what the pink colour with phenolphthalein showed. (b) Which chemical family does sodium belong to, and how many outer electrons does each atom have? Write its electron arrangement. (c) Explain why sodium reacts so violently with water. (d) Write the balanced equation for the reaction. (e) Lithium, sodium and potassium are in the same group. Explain how their reactivity compares and why.

(a) The sodium floated, fizzed, darted about and melted into a ball. The solution turned pink with phenolphthalein, which showed that an alkali (sodium hydroxide, NaOH) formed. (b) Sodium is an alkali metal (Group 1). Each atom has 1 outer electron. Its electron arrangement is 2,8,1. (c) Claim: Sodium is violent because its single outer electron is lost very easily. Evidence: it reacts fast in cold water, gives off hydrogen gas and melts. Reasoning: an atom that loses its outer electron easily reacts readily, so sodium is dangerous on its own and must be stored under paraffin oil and handled only by the teacher. (d) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$. (e) Reactivity increases down the group: $\text{Li} < \text{Na} < \text{K}$. Lithium fizzed slowly, sodium fast, and potassium very fast with a lilac flame. Down the group the outer electron is further from the nucleus, so it is lost more easily and reactivity increases.

Part 2: Sodium, chlorine and safe salt, and chlorine in Mombasa water

Mama Wanjiru asks why chlorine in her tap water is safe when chlorine as a gas is poisonous, and why the sodium that is dangerous alone is in the salt she puts in her food. (a) Which chemical family is chlorine in, and how many outer electrons does it have? Write its electron arrangement. (b) Write the balanced equation for sodium reacting with chlorine. (c) Use the idea of outer electrons and what you know about the dose to explain to Mama Wanjiru why the tap water is safe. (d) Use outer electrons to explain why table salt is safe to eat even though sodium alone is dangerous.	(a) Chlorine is a halogen (Group 17). It has 7 outer electrons. Its electron arrangement is 2,8,7. It is a reactive non-metal. (b) $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$. (c) Claim: the water is safe because only a tiny dose is used. Evidence: chlorine is added at about 0.5 mg/L at the Mombasa water works, which means only 10 mg in 20 L of water, and at this dose it kills the bacteria that cause cholera and typhoid. Reasoning: a chlorine atom has 7 outer electrons and needs just one more, so it reacts readily, which is why it is good at killing microbes. Because it is so reactive it is toxic in large amounts, so only a small safe dose is used. (d) Sodium has 1 outer electron that it loses very easily, and chlorine needs one more outer electron to have 8. When they react, the outer electron of sodium is taken up by chlorine and they form the compound sodium chloride (NaCl), which is stable and has completely different properties from its elements. Sodium no longer has its easily lost outer electron to react violently with water, and chlorine is no longer a reactive poisonous gas, because both are combined in the stable compound, so the salt is safe to eat.

Part 3: Reactivity and position in the periodic table	
Akinyi asked, 'If sodium is so dangerous, are all metals like that?' (a) Describe what happened when you tested magnesium ribbon in cold water, in dilute HCl and in air. (b) Compare magnesium with sodium using their electron arrangements and the evidence. (c) Write the balanced equations for magnesium burning in air and for magnesium reacting with dilute HCl. (d) Neon and argon do not react at all. Using outer electrons, explain why, and state a use that depends on this.	(a) Magnesium showed no visible reaction in cold water. It gave steady bubbles of hydrogen gas in dilute HCl (the pop test was positive). It burned with a bright white flame in air, leaving white MgO powder. (b) Sodium is 2,8,1 (1 outer electron) and magnesium is 2,8,2 (2 outer electrons, an alkaline earth metal in Group 2). Magnesium has to lose 2 outer electrons rather than 1, so it loses them less readily than sodium does. This matches the evidence: magnesium shows no reaction in cold water, whereas sodium fizzes fast and melts. So magnesium is less reactive than sodium, and the reactivity order is $\text{K} > \text{Na} > \text{Li} > \text{Ca} > \text{Mg}$. So not all metals are as dangerous as sodium. Position in the periodic table helps us predict this: Group 2 is less reactive than Group 1. (c) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$. (d) Noble gases (Group 18) have full outer shells (Ne 2,8; Ar 2,8,8), so their atoms do not need to lose or gain electrons and they are unreactive. Because neon is unreactive, it suits signs.

Part 4: Matching metals to jobs

Use the data given in the instructions and what you learned about the metals in the puzzle. (a) Using the data, give reasons why aluminium is used for Mama Wanjiru's sufuria. (b) Why is copper used for the wire in a kettle or jiko cable? (c) Why is the mabati roof made of iron coated with zinc? (d) Lead is the densest metal in the data but its use is being reduced. Explain why, using what you learned. (e) Compare sodium with aluminium using the data and explain why sodium would not suit a sufuria.

(a) Aluminium is light (2.70 g/cm^3 , much less dense than copper at 8.96 g/cm^3), so the pot is easy to carry. It is a metal that conducts, so heat from the fire passes through it to the food, and it melts at 660 C , well above cooking temperatures, so it stays solid on the fire. It also resists corrosion because of its protective oxide layer, so it stays in good condition and is safe for cooking, unlike sodium. (b) Copper is an excellent conductor of electricity and can be drawn into wire (ductile). (c) Iron is strong and cheap, so it forms the base of the mabati. Zinc is more reactive than iron, so it corrodes first and protects the iron from rusting (sacrificial protection). (d) Lead is very dense (11.34 g/cm^3) and is used in car batteries and radiation shielding, but it is toxic, so its use is being reduced. This shows that elements are chosen for their properties but with trade-offs such as safety. (e) Sodium is less dense than water (0.97 g/cm^3 compared with 1.00 g/cm^3) so it floats, and it melts at only 98 C , while aluminium melts at 660 C . Sodium also reacts violently with water. A sodium pot would melt on a fire and be dangerous, so aluminium is the better choice.

Part 5: Answering the driving question

Driving question: Why does each element behave and get used the way it does, and how can the periodic table help us predict which element is right for which job in Kenyan daily life? (a) Write a final explanation of the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle using claim, evidence and reasoning. Include outer electrons and position in the periodic table. (b) A jua kali artisan in Gikomba wants to make a new product: a cooking pot, a cable or a roof sheet. Choose one product, recommend an element and justify your choice using properties, safety, cost and availability. (c) How has your thinking changed since your first model in Lesson 1?

(a) Claim: Each element is used for a job because of its properties, which depend on its outer electrons and so on its position in the periodic table. Evidence: sodium (Group 1, 1 outer electron) reacts violently with water, but as NaCl it is stable and safe to eat. Chlorine (Group 17, 7 outer electrons) is a toxic gas in large amounts but kills microbes at about 0.5 mg/L in Mombasa water. Aluminium is light and conducts heat, so it suits the sufuria. Copper is an excellent conductor, so it suits wire. Zinc coats iron on mabati. Neon (full outer shell) is unreactive, so it suits signs, and sodium vapour gives the yellow-orange light of the Thika Road lamps. Reasoning: the periodic table lets us predict family behaviour, such as reactivity rising down Group 1, and so match element to job while balancing safety, cost and availability. (b) For a roof sheet I would recommend iron coated with zinc (galvanised mabati). Iron is strong and cheap, and zinc protects it from rusting. Both are available to Gikomba artisans, and the metals are safe to handle, unlike lead, which is toxic. (c) At first I thought each element is simply used for what it is good at, and I could not explain why sodium is dangerous but salt is safe. Now I can explain with outer electrons and position in the periodic table that reactivity and properties differ between families, and that compounds have different properties from their elements.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of outer electrons and periodic table position	Correctly links group, outer electrons, electron arrangements and reactivity for Na, Mg, Cl and noble gases, including the trend down Group 1.	Gives correct outer electron numbers and links most to reactivity, with minor gaps in the reasoning.	Gives few outer electron facts or does not link them to behaviour or position.
Use of evidence from the lessons	Uses accurate observations and data (sodium demo, Mg tests, 0.5 mg/L, densities and melting points) to support every claim.	Uses some accurate evidence but misses some data or gives it loosely.	Gives little evidence, or the data is inaccurate.
Chemical equations and terms	All equations are balanced and correct; element, compound, group and chemical family are used correctly.	Most equations are correct with one small error; terms mostly correct.	Equations are unbalanced or missing; terms are confused.
Matching element to job and trade-offs	Explains each job by properties and gives a justified recommendation weighing safety, cost and availability, including lead's toxicity.	Matches properties to jobs but gives limited discussion of trade-offs.	States uses without linking them to properties.
Claim-evidence-reasoning and explaining the puzzle	Gives a clear claim, evidence and reasoning that fully explains why sodium is dangerous alone but safe as salt, and why chlorine is safe at a tiny dose.	Explains most of the puzzle, with some reasoning missing.	Gives a partial or unclear explanation of the puzzle.

ATTRIBUTION AND LICENSING

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